

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Zoology

Paper: ZOB-602 — Mammalian Endocrinology

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False with reason in not more than 2-3 lines each: [7 × 2 = 14]

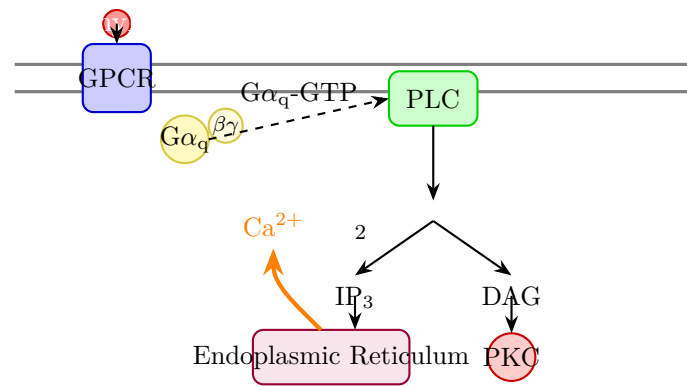
- (i) Higher dopamine level suppresses milk release from lactating female.
- (ii) SCN, SON, ARC and PVN neurons are located in tuberal region of hypothalamus.
- (iii) In the pituitary, TRH-R2 mediates the TRH signal.
- (iv) Relaxin inhibits matrix metalloproteinases and collagenase activity of the pubic symphysis peripartum.
- (v) Lutein cells show higher CYP 19 activity as compared to dominant follicular granulosa cells.
- (vi) Oxytocin exerts myotropic action on the uterus.
- (vii) Acidosis is one of the factors responsible for hypocalcemic state.

(b) Define the following: [6 × 1 = 6]

- (i) Pentagastrin
- (ii) KNDy neurons
- (iii) PNMT
- (iv) Circadian rhythms
- (v) Juxtaglomerular apparatus
- (vi) Phosphodiesterase.

Question 2. The androgen receptor is found in almost every tissue in primates, not just the classical androgen-dependent organs. As a result, androgens have a wide range of effects on a wide range of targets. Discuss. [10]

Question 3. Describe in detail the on and off mechanisms of $G\alpha_q$ signalling pathway. [10]

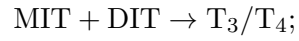


Question 4. Explain the biosynthesis of mineralocorticoids. Comment upon the functions of aldosterone and regulation of its release. [10]

Question 5. Write short notes on the following: [2 × 5 = 10]

(a) Goitre

(b) Biosynthesis of thyroid hormone



[->, thick] (2.8, -1.8) to[out=180, in=-45] (0.5, -2.2) node[above left, font=] Endocytosis; [->,

thick] (-0.2, -2.2) - (-3.8, -2.2) node[left] T₃, T₄;

Question 7. Describe the biosynthesis and biological actions of melatonin.

[10]

Serotonin N-acetyltransferase
 Tryptophan 5-hydroxylase
 (Acetyl CoA MT)

Question 8. A person is diagnosed with hyperglycemia, glycosuria and polydipsia. Give an account of the hormonal dysfunction and mechanism of pathophysiological consequences.

[10]

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